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Steerable crossing catheter

Abstract

Medical devices as well as methods for making and using medical devices are disclosed. An example medical device may include a steerable crossing catheter system. The steerable crossing catheter system may include an elongate catheter shaft having a proximal end region, a steerable distal end region, and a lumen extending therethrough. A core may be disposed within the lumen. The core may have a distal end. A penetrating member may be coupled to the distal end. A sensing assembly may be disposed adjacent to the distal end. The sensing assembly may be configured to estimate the location of the core within a body lumen.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/165,182, filed on Feb. 2, 2021, which claims the benefit of and priority to U.S. Provisional Patent Application Ser. No. 62/969,252 filed on Feb. 3, 2020, the disclosures of which are incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure pertains to medical devices, and methods for manufacturing medical devices. More particularly, the present disclosure pertains to medical devices for crossing vascular occlusions.

BACKGROUND

(2) A wide variety of intracorporeal medical devices have been developed for medical use, for example, intravascular use. Some of these devices include guidewires, catheters, and the like. These devices are manufactured by any one of a variety of different manufacturing methods and may be used according to any one of a variety of methods. Of the known medical devices and methods, each has certain advantages and disadvantages. There is an ongoing need to provide alternative medical devices as well as alternative methods for manufacturing and using medical devices.

BRIEF SUMMARY

(3) This disclosure provides design, material, manufacturing method, and use alternatives for medical devices. A steerable crossing catheter system is disclosed. The steerable crossing catheter system comprises: an elongate catheter shaft having a proximal end region, a steerable distal end region, and a lumen extending therethrough; a core disposed within the lumen, the core having a distal end; a penetrating member coupled to the distal end; and a sensing assembly disposed adjacent to the distal end, the sensing assembly being configured to estimate the location of the

core within a body lumen.

(4) Alternatively or additionally to any of the embodiments above, the steerable distal end region includes a steering member.

(5) Alternatively or additionally to any of the embodiments above, the steering member includes a tubular member having a plurality of slots formed therein.

(6) Alternatively or additionally to any of the embodiments above, further comprising an actuator coupled to the steering member.

(7) Alternatively or additionally to any of the embodiments above, the core is slidably disposed within the lumen.

(8) Alternatively or additionally to any of the embodiments above, the sensing assembly includes an ultrasound transducer.

(9) Alternatively or additionally to any of the embodiments above, the penetrating member includes a pointed tip at the distal end of the core.

(10) Alternatively or additionally to any of the embodiments above, the penetrating member includes an angled region at the distal end of the core.

(11) A method for crossing a venous occlusion is disclosed. The method comprises: advancing a medical device system through a vein to a position adjacent to a venous occlusion; wherein the medical device system includes a steering member, a penetrating tip, and a sensing assembly; advancing the penetrating tip into engagement with the venous occlusion; estimating the position of the medical device system within the vein using the sensing assembly; and steering the medical device system with the steering member.

(12) Alternatively or additionally to any of the embodiments above, the medical device system includes a catheter and a core disposed within the catheter.

(13) Alternatively or additionally to any of the embodiments above, the steering member is disposed along a distal end region of the catheter.

(14) Alternatively or additionally to any of the embodiments above, the penetrating tip is disposed adjacent to a distal end of the core.

(15) Alternatively or additionally to any of the embodiments above, the sensing assembly is disposed adjacent to a distal end of the core.

(16) Alternatively or additionally to any of the embodiments above, the sensing assembly includes an ultrasound transducer.

(17) Alternatively or additionally to any of the embodiments above, estimating the position of the medical device system within the vein using the sensing assembly includes emitting ultrasonic energy with the ultrasound transducer.

(18) Alternatively or additionally to any of the embodiments above, steering the medical device system with the steering member includes steering the medical device system so that the penetrating tip avoids contact with the vein.

(19) A medical device system is disclosed. The medical device system comprises: a catheter; a steering member coupled to the catheter; a core disposed within the catheter; a penetrating member coupled to the core; and a sensing assembly coupled to the core, the sensing assembly being configured to estimate the location of the core within a body lumen during a medical intervention to cross a venous occlusion.

(20) Alternatively or additionally to any of the embodiments above, the steering member includes a tubular member having a plurality of slots formed therein.

(21) Alternatively or additionally to any of the embodiments above, further comprising an actuator coupled to the steering member.

(22) Alternatively or additionally to any of the embodiments above, the sensing assembly includes an ultrasound transducer.

(23) The above summary of some embodiments is not intended to describe each disclosed

embodiment or every implementation of the present disclosure. The Figures, and Detailed Description, which follow, more particularly exemplify these embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The disclosure may be more completely understood in consideration of the following detailed description in connection with the accompanying drawings, in which:
- (2) FIG. 1 is a side view of an example medical device assembly.
- (3) FIG. 2 is a side view of an example medical device.
- (4) FIG. 3 is a side view of an example medical device.
- (5) FIGS. 4-6 illustrate an example use of an example medical device assembly.
- (6) FIGS. 7-11 illustrate example penetrating tips.
- (7) FIG. 12 is a side view of an example medical device.
- (8) FIG. 13 is a side view of an example medical device.
- (9) While the disclosure is amenable to various modifications and alternative forms, specifics thereof have been shown by way of example in the drawings and will be described in detail. It should be understood, however, that the intention is not to limit the invention to the particular embodiments described. On the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the disclosure.

DETAILED DESCRIPTION

- (10) For the following defined terms, these definitions shall be applied, unless a different definition is given in the claims or elsewhere in this specification.
- (11) All numeric values are herein assumed to be modified by the term “about”, whether or not explicitly indicated. The term “about” generally refers to a range of numbers that one of skill in the art would consider equivalent to the recited value (e.g., having the same function or result). In many instances, the terms “about” may include numbers that are rounded to the nearest significant figure.
- (12) The recitation of numerical ranges by endpoints includes all numbers within that range (e.g. 1 to 5 includes 1, 1.5, 2, 2.75, 3, 3.80, 4, and 5).
- (13) As used in this specification and the appended claims, the singular forms “a”, “an”, and “the” include plural referents unless the content clearly dictates otherwise. As used in this specification and the appended claims, the term “or” is generally employed in its sense including “and/or” unless the content clearly dictates otherwise.
- (14) It is noted that references in the specification to “an embodiment”, “some embodiments”, “other embodiments”, etc., indicate that the embodiment described may include one or more particular features, structures, and/or characteristics. However, such recitations do not necessarily mean that all embodiments include the particular features, structures, and/or characteristics. Additionally, when particular features, structures, and/or characteristics are described in connection with one embodiment, it should be understood that such features, structures, and/or characteristics may also be used connection with other embodiments whether or not explicitly described unless clearly stated to the contrary.
- (15) The following detailed description should be read with reference to the drawings in which similar elements in different drawings are numbered the same. The drawings, which are not necessarily to scale, depict illustrative embodiments and are not intended to limit the scope of the invention.
- (16) Vascular occlusions can result in serious health complications. For example, occlusions along the venous system can result in the pooling of blood in the lower extremities, leg swelling, skin changes, pain, ulcers, and/or the like. Treating venous occlusions generally includes the restoration

of venous outflow back to the heart. Some example treatments may include angioplasty, stenting, and/or the like. In order to implement such treatments, it may be useful to pass a device such as a catheter and/or a guidewire through the occlusion. Unlike arterial occlusions that tend to become calcified, venous occlusions may have a tendency to become tough or hardened more like scar tissue. Because of this, the occlusion may tend to blend together with the vessel wall such that attempting to cross the occlusion with a medical device could bring the medical device into close proximity with the vessel wall. In some cases, attempts to cross the occlusion could result in damage to the vessel wall. Disclosed herein are medical device systems that can be used to cross an intravascular (e.g., venous) occlusion such a chronic total occlusion (e.g., a chronic total venous occlusion). Some of the features of these medical device systems are disclosed herein.

(17) FIG. 1 schematically depicts an example medical device system **10**. The medical device system **10** includes a shaft or catheter **12** and a core **14**. In some instances, the core **14** is secured to and/or fixed to the catheter **12**. In other instances, the core **14** is free from direct attachment to the catheter **12**. For example, the core **14** may be slidably disposed within the lumen of the catheter **12**. In such instances, the core **14** may be disposed within the catheter **12** during a crossing procedure. At a suitable time during a crossing procedure, the core **14** can be removed from the catheter **12**. This may allow a guidewire to be passed through the catheter **12**, another device such as a treatment device to be passed through the catheter **12**, another “core” device to be passed through the catheter **12**, etc.

(18) FIG. 2 illustrates the catheter **12**. In at least some instances, the catheter **12** may be similar in form and function to a guide catheter. The catheter **12** may include a proximal portion **16** and a distal portion **18**. Structurally, the catheter **12** may include one or more layers of material. For example, the catheter **12** may include an inner layer or liner, which may be formed from a lubricious material. In some instances, a reinforcement layer may be disposed along the inner liner. The reinforcement layer may include a braid, coil, and/or the like. In some instances, an outer layer or sleeve may be disposed along the reinforcement layer. The outer layer may have a constant shore hardness or stiffness along its length or may include sections with differing levels of hardness/stiffness. This is just one example construction of the catheter **12**. Other constructions are contemplated.

(19) In some instances, the catheter **12** may include one or more features that allow the catheter **12**, for example the distal portion **18** of the catheter **12**, to be steerable. For example, the catheter **12** may include a steering member **20**. In this example, the steering member **20** takes the form of a tube or tubular member having a plurality of slots **22** formed therein. The slots **22** may be disposed along one or more sides of the steering member **20** so that the steering member **20** may have one or more (e.g., one, two, three, four, or more) preferred bending directions. One or more actuators **24** may be coupled to the steering member **20**. In this example, the actuator **24** may take the form of a pull wire coupled to the steering member **20** (e.g., adjacent to a distal end of the steering member **20**) and extending proximally therefrom. The actuator **24** may allow a clinician to cause the catheter **12** to bend, curve, or otherwise change in shape. Thus, a clinician can use the actuator **24** to help navigate or steer the catheter **12** during an intravascular procedure. Some examples of suitable steering members and/or structures that may be incorporated into the catheter **12** in order to make the catheter **12** steerable include those disclosed in U.S. Pat. No. 9,162,046, the entire disclosure of which is herein incorporated by reference.

(20) FIG. 3 illustrates the core **14**. The core **14** may take the form of a shaft **25** having a proximal end region **26** and a distal end region **28**. In some instances, the core **14** may include a guidewire lumen (not shown) generally extending at least partially therethrough. The core **14** includes a number of structural features. For example, the core **14** may have a penetrating member or tip **30** coupled to the distal end region **28**. The penetrating tip **30** may be generally configured to aid in engaging, penetrating, and ultimately crossing an intravascular occlusion. As such, the penetrating tip **30** may have a shape or configuration that could be described as being sharp or sharpened,

pointed, and/or the like.

(21) The core **14** may also include a sensing and/or imaging assembly **32**. The form of the sensing and/or imaging assembly **32** may vary. For the purposes of this disclosure, the terms “sensing and/or imaging” may be understood to mean that structures typically used for sensing as well as structures typically used for imaging can make up parts of the sensing and/or imaging assembly **32** and such structures can be used to gather/collect information useful for aiding a clinician during a crossing procedure. For example, the sensing and/or imaging assembly **32** may take the form of an intravascular ultrasound (IVUS) assembly and/or at least one ultrasound transducer. An ultrasound transducer, for example, can be used to emit and measure ultrasonic reflections from within the vessel in order to estimate the tissue properties adjacent to the core **14** and, ultimately, use these properties to estimate the location of the core **14** within the vessel. For example, an occlusion and/or the tissue adjacent to the occlusion may include collagen. The type of collagen (e.g., type I vs. type III collagen) can vary through an occlusion (e.g., where the concentration of type III collagen tends to increase from the center of an occlusion toward the edge). Surrounding tissue and/or the vessel wall, conversely, may have a higher concentration of type I collagen. Because type I collagen and type III collagen reflect ultrasonic energy differently, a clinician can use the sensing and/or imaging assembly **32** to estimate the relative location of the core **14** within the vessel by sensing/imaging the different types of collagen adjacent to the core **14**. More particularly, the sensing and/or imaging assembly **32** can be used to determine whether or not the core **14** is approaching tissue (e.g., the vessel wall) during a crossing procedure by sensing/imaging the type of collagen adjacent to the sensing and/or imaging assembly **32** within the vessel.

(22) In some crossing procedures, a stent may already be in place adjacent to the occlusion. The sensing and/or imaging assembly **32** can also be used to sense/visualize the stent in order to help cross the occlusion. In such instances, the stent may provide another indicator of the location of the vessel wall that can be used to help reduce the likelihood of engaging/damaging the vessel wall.

(23) In instances where the sensing and/or imaging assembly **32** includes an ultrasound transducer, the core **14** may include a number of structural features commonly used with ultrasound and/or IVUS devices. For example, the sensing and/or imaging assembly **32** may include a sensing/imaging device or transducer **36** that is coupled to a drive cable **34**. For example, the sensing/imaging transducer **36** may include an ultrasound transducer or an array of (e.g., two or more) ultrasound transducers. One or more conductive member **38** may be coupled to the sensing/imaging transducer **36**. The core **14** may also include a motor **40**. The motor **40** may include a magnet **42** that can be driven to rotate by one or more magnetic field windings **44**. One or more current lines **46** may be coupled to the windings **44**. The motor **40** can be utilized to rotate the sensing and/or imaging assembly **32**. These structures are just examples. Additional/different structures may be utilized. For example, in some instances, the sensing/imaging transducer **36** can be rotated by rotating the drive cable **34**. In some of these and in other instances, the core **14** can be rotated in order to point the sensing/imaging transducer **36** in a variety of different directions.

(24) In some instances, the sensing/imaging transducer **36** generally points to or is oriented in a direction substantially normal to the longitudinal axis of the core **14**. In other words, the sensing/imaging transducer **36** may be pointed to the side or otherwise configured to sense/image objects alongside the core **14**. In such instances, rotation of the sensing and/or imaging assembly **32** (and/or the sensing/imaging transducer **36**), for example by the motor **40** or by rotating the drive cable **34**, allows for sensing/imaging data to be gathered/collected from a plurality of different vantage points about the core **14**. In other instances, the sensing/imaging transducer **36** may be pointed in the longitude direction. In some of these instances, the sensing/imaging transducer **36** may interact with other structures such as mirror and/or mirror holder (not shown). In such instances, the motor **40** may be used to rotate the mirror so that sensing/imaging data can be gathered/collected from a plurality of different vantage points about the core **14**. Some examples of sensing and/or imaging assemblies and/or sensing/imaging transducers **36** that may be utilized with

the core **14** are disclosed in U.S. Patent Application Pub. No. US 2012/0059241, the entire disclosure of which is herein incorporated by reference.

(25) While the sensing/imaging transducer **36** may take the form of or otherwise include an ultrasound transducer, this is not intended to be limiting. For example, the sensing and/or imaging assembly **32** may include different structural features including, but not limited to, an impedance sensor and/or electrode, an optical system/sensor, a biochemical system/sensor, a mechanical/piezo-electric sensor, combinations thereof, and/or the like. Such structures may be used similarly to the sensing and/or imaging assembly **32** in order to estimate the position of the core **14** within the vessel. For example, when using an impedance sensor (e.g., which may be disposed along the core **14**, the catheter **12**, or both) a venous occlusion may be more electrically resistive than surrounding tissue. Thus, an electrical pulse from an impedance electrode to a grounding pad disposed on the patient may provide an impedance/resistance value/profile that can be used to estimate the position of the core **14** within the vessel. When using an optical system, fluorescence spectroscopy can be used to characterize tissue. For example, different types of collagen may reflect light differently such that processing the reflected light (e.g., via a reflected wavelength pattern, a reflected light intensity, combinations thereof, etc.) may be used to estimate the position of the core **14** within the vessel. When using a biochemical sensor, a probe may be coated, for example, with lysyl oxidase, which may interact with lysine (e.g., which is a building block of collagen) and create an electrical signal that can be measured and translated into collagen concentration and, thus, estimate the position of the core **14** within the vessel. When using a mechanical/piezo-electric sensor, an electrical signal may be used in response to mechanical pressure changes during engagement with occlusions. Such sensing can be used estimate the position of the core **14** within the vessel.

(26) While the above discussion is directed to a sensing and/or imaging assembly **32** that is a component of the core **14**, other medical device systems are contemplated that either additionally or alternatively incorporate a sensing and/or imaging assembly into the catheter **12**. For example, an ultrasound transducer can be disposed along the distal portion **18** of the catheter **12**. Such an ultrasound transducer can be used analogously to the sensing and/or imaging assembly **32** described herein. Alternative sensing and/or imaging assemblies can be incorporated into the catheter **12**.

(27) Collectively, the structural features of the catheter **12** and the core **14** may allow the medical device system **10** to efficiently cross an intravascular occlusion (e.g., a venous occlusion). For example, the medical device system **10** includes the ability to engage and/or pierce into an occlusion (e.g., via the penetrating tip **30**), estimate the position of the medical device system **10** within the vessel (e.g., via sensing and/or imaging assembly **32**), and steer the medical device system **10** (e.g., via the steering member **20**) so that contact with or damage to the vessel wall can be minimized/avoided during a crossing procedure.

(28) FIGS. 4-6 depict an example use of the medical device system **10** during a crossing procedure. Here it can be seen how the structural features of the catheter **12** and the core **14** desirably impact the crossing procedure. For example, the medical device system **10** may be navigated through a blood vessel **48** to a position adjacent to an occlusion **50**. When doing so, the core **14** may be disposed within the catheter **12** such that the penetrating tip **30** extends distally from the distal portion **18** of the catheter **12** during navigation. Upon approaching the occlusion **50**, the penetrating tip **30** may engage the occlusion **50** as shown in FIG. 4. When doing so, the penetrating tip **30** may begin to pierce through the occlusion **50** such that the medical device system **10** begins to cross the occlusion **50**. While passing through the occlusion **50**, the sensing and/or imaging assembly **32** can be utilized in order to estimate the position of the medical device system **10** within the vessel **48** and to monitor whether or not the medical device system **10** is approaching the wall of the vessel **48**. At some point during the crossing procedure, it may be desirable to steer the catheter **12** in order to advance the medical device system **10** through the occlusion **50**. For example, the penetrating tip **30** may encounter a region of the occlusion **50** that is not easily passed (e.g., the

occlusion **50** is tough or hardened) or the penetrating tip **30** may begin to approach the wall of the vessel **48**. When this happens, a clinician may utilize the steering member **20** to bend or steer the distal portion **18** of the catheter **12** away from the toughened region and/or away from the wall of the vessel **48**. The steering of the medical device system **10** is schematically depicted in FIG. **5**. The crossing procedure may include additional monitoring of the position of the medical device system **10** within the vessel **48** and/or one or more additional steering processes until the medical device system **10** is able to successfully cross the occlusion **50** as depicted in FIG. **6**. In some instances, when the medical device system **10** has crossed the occlusion **50**, the core **14** can be removed from the catheter **12**. Once the core **14** is removed, a guidewire, treatment device, or another medical device can be passed through the catheter **12** in order to implement a suitable treatment for the occlusion **50**.

(29) FIGS. **7-11** illustrate a number of alternative penetrating members/tips that may have similar form and function to the penetrating tip **30**. As such, these alternative penetrating members can be incorporated into the medical device system **10**. For example, FIG. **7** illustrates a penetrating tip **130** that includes an angled region **152**. FIG. **8** illustrates a penetrating tip **230** with a conical region **252**. FIG. **9** illustrates a penetrating tip **330** with a tapering region **352**. FIG. **10** illustrates a penetrating tip **430** with a plurality of angled regions **452**. FIG. **11** illustrates a penetrating tip **530** that is slightly curved and that includes an angled region **552**. These penetrating tips are meant to be examples. Additional penetrating tips are contemplated.

(30) FIG. **12** illustrates another example core **614** that may be similar in form and function to other cores disclosed herein. The core **614** may include a shaft **625** having a penetrating tip **630** coupled thereto. The core **614** may include a sensing and/or imaging assembly **632**. The sensing and/or imaging assembly **632** may include an optical fiber **654**. As discussed briefly herein, the optical fiber **654** may be used to shine light into the vessel. In order for the light to reach the tissue/occlusion, the penetrating tip **630** may be substantially transparent. Light that is reflected back can be used to characterize the tissue and/or estimate the position of the core **14** within the vessel. For example, different types of collagen may reflect light differently such that processing the reflected light (e.g., via a reflected wavelength pattern, a reflected light intensity, combinations thereof, etc. using fluorescence spectroscopy) may be used to estimate the position of the core **14** within the vessel.

(31) FIG. **13** illustrates another example core **714** that may be similar in form and function to other cores disclosed herein. The core **714** may include a shaft **725** having a penetrating tip **730** coupled thereto. The core **714** may include a sensing and/or imaging assembly **732**. The sensing and/or imaging assembly **732** may include an optical fiber **754** and a reflective member **756**. In this example, a portion of shaft **725** may be substantially transparent so that light emitted from the optical fiber **754** can reach the tissue/occlusion. The reflective member **756** may help direct the light to transparent regions of the shaft **725**.

(32) The materials that can be used for the various components of the medical device system **10** may include those commonly associated with medical devices. For simplicity purposes, the following discussion makes reference to the catheter **12**. However, this is not intended to limit the devices and methods described herein, as the discussion may be applied to other structures or devices disclosed herein.

(33) The catheter **12** may be made from a metal, metal alloy, polymer (some examples of which are disclosed below), a metal-polymer composite, ceramics, combinations thereof, and the like, or other suitable material. Some examples of suitable polymers may include polytetrafluoroethylene (PTFE), ethylene tetrafluoroethylene (ETFE), fluorinated ethylene propylene (FEP), polyoxymethylene (POM, for example, DELRIN® available from DuPont), polyether block ester, polyurethane (for example, Polyurethane 85A), polypropylene (PP), polyvinylchloride (PVC), polyether-ester (for example, ARNITEL® available from DSM Engineering Plastics), ether or ester based copolymers (for example, butylene/poly(alkylene ether) phthalate and/or other polyester

elastomers such as HYTREL® available from DuPont), polyamide (for example, DURETHAN® available from Bayer or CRISTAMID® available from Elf Atochem), elastomeric polyamides, block polyamide/ethers, polyether block amide (PEBA, for example available under the trade name PEBAX®), ethylene vinyl acetate copolymers (EVA), silicones, polyethylene (PE), MARLEX® high-density polyethylene, MARLEX® low-density polyethylene, linear low density polyethylene (for example REXELL®), polyester, polybutylene terephthalate (PBT), polyethylene terephthalate (PET), polytrimethylene terephthalate, polyethylene naphthalate (PEN), polyetheretherketone (PEEK), polyimide (PI), polyetherimide (PEI), polyphenylene sulfide (PPS), polyphenylene oxide (PPO), poly paraphenylene terephthalamide (for example, KEVLAR®), polysulfone, nylon, nylon-12 (such as GRILAMID® available from EMS American Grilon), perfluoro(propyl vinyl ether) (PFA), ethylene vinyl alcohol, polyolefin, polystyrene, epoxy, polyvinylidene chloride (PVdC), poly(styrene-b-isobutylene-b-styrene) (for example, SIBS and/or SIBS 50A), polycarbonates, ionomers, biocompatible polymers, other suitable materials, or mixtures, combinations, copolymers thereof, polymer/metal composites, and the like. In some embodiments the sheath can be blended with a liquid crystal polymer (LCP). For example, the mixture can contain up to about 6 percent LCP.

(34) Some examples of suitable metals and metal alloys include stainless steel, such as 304V, 304L, and 316LV stainless steel; mild steel; nickel-titanium alloy such as linear-elastic and/or super-elastic nitinol; other nickel alloys such as nickel-chromium-molybdenum alloys (e.g., UNS: N06625 such as INCONEL® 625, UNS: N06022 such as HASTELLOY® C-22®, UNS: N10276 such as HASTELLOY® C276®, other HASTELLOY® alloys, and the like), nickel-copper alloys (e.g., UNS: N04400 such as MONEL® 400, NICKELVAC® 400, NICORROS® 400, and the like), nickel-cobalt-chromium-molybdenum alloys (e.g., UNS: R30035 such as MP35-N® and the like), nickel-molybdenum alloys (e.g., UNS: N10665 such as HASTELLOY® ALLOY B2®), other nickel-chromium alloys, other nickel-molybdenum alloys, other nickel-cobalt alloys, other nickel-iron alloys, other nickel-copper alloys, other nickel-tungsten or tungsten alloys, and the like; cobalt-chromium alloys; cobalt-chromium-molybdenum alloys (e.g., UNS: R30003 such as ELGILOY®, PHYNOX®, and the like); platinum enriched stainless steel; titanium; combinations thereof; and the like; or any other suitable material.

(35) In at least some embodiments, portions or all of the medical device system **10** may also be doped with, made of, or otherwise include a radiopaque material. Radiopaque materials are understood to be materials capable of producing a relatively bright image on a fluoroscopy screen or another imaging technique during a medical procedure. This relatively bright image aids the user of the medical device system **10** in determining its location. Some examples of radiopaque materials can include, but are not limited to, gold, platinum, palladium, tantalum, tungsten alloy, polymer material loaded with a radiopaque filler, and the like. Additionally, other radiopaque marker bands and/or coils may also be incorporated into the design of the medical device system **10** to achieve the same result.

(36) In some embodiments, a degree of Magnetic Resonance Imaging (MRI) compatibility is imparted into the medical device system **10**. For example, the medical device system **10**, or portions thereof, may be made of a material that does not substantially distort the image and create substantial artifacts (e.g., gaps in the image). Certain ferromagnetic materials, for example, may not be suitable because they may create artifacts in an MRI image. The medical device system **10**, or portions thereof, may also be made from a material that the MRI machine can image. Some materials that exhibit these characteristics include, for example, tungsten, cobalt-chromium-molybdenum alloys (e.g., UNS: R30003 such as ELGILOY®, PHYNOX®, and the like), nickel-cobalt-chromium-molybdenum alloys (e.g., UNS: R30035 such as MP35-NR and the like), nitinol, and the like, and others.

(37) It should be understood that this disclosure is, in many respects, only illustrative. Changes may be made in details, particularly in matters of shape, size, and arrangement of steps without

exceeding the scope of the disclosure. This may include, to the extent that it is appropriate, the use of any of the features of one example embodiment being used in other embodiments. The invention's scope is, of course, defined in the language in which the appended claims are expressed.

Claims

1. A method comprising: advancing a medical device through a vessel, wherein the medical device includes a steering member, a penetrating tip, a core, and a sensing assembly coupled to the core; sensing tissue properties adjacent to the sensing assembly within the vessel; estimating a position of the medical device within the vessel based on the tissue properties; and steering the medical device with the steering member based on the estimated position; wherein the tissue properties include different types of collagen.
 2. The method of claim 1, wherein the sensing assembly includes an ultrasound transducer, and further comprising emitting ultrasonic energy with the ultrasound transducer to sense the tissue properties.
 3. The method of claim 1, wherein the sensing assembly includes an optical fiber, and further comprising emitting light with the optical fiber to sense the tissue properties.
 4. The method of claim 3, further comprising sensing the tissue properties with fluorescence spectroscopy.
 5. The method of claim 1, wherein the vessel is a blood vessel.
 6. The method of claim 5, wherein the vessel is a vein.
 7. The method of claim 5, wherein steering the medical device with the steering member includes steering the medical device so that the penetrating tip avoids contact with the blood vessel.
 8. The method of claim 1, further comprising advancing the medical device through the vessel to a position adjacent to an occlusion.
 9. The method of claim 8, further comprising advancing the penetrating tip into engagement with the occlusion.
 10. The method of claim 1, wherein the medical device includes a catheter, and wherein the core is disposed within the catheter.
 11. The method of claim 10, wherein the steering member is disposed along a distal end region of the catheter, and wherein the penetrating tip is disposed adjacent to a distal end of the core.
 12. The method of claim 1, further comprising estimating a position of the core within the vessel based on the tissue properties.
 13. A method comprising: advancing a medical device through a blood vessel, wherein the medical device includes a steering member, a penetrating tip, a core, and a sensing assembly coupled to the core; sensing tissue properties adjacent to the sensing assembly within the blood vessel; estimating a position of the medical device within the blood vessel based on the tissue properties; and steering the medical device with the steering member based on the estimated position; wherein the tissue properties include concentrations of different types of collagen.
 14. A method comprising: advancing a medical device through a blood vessel to a position adjacent to an occlusion, wherein the medical device includes a steering member, a core, a penetrating tip coupled to the core, and a sensing assembly coupled to the core; emitting ultrasonic energy to sense tissue properties adjacent to the sensing assembly within the vessel; estimating a position of the core within the blood vessel based on the tissue properties; and steering the medical device with the steering member based on the estimated position of the medical device within the blood vessel; wherein the tissue properties include different types of collagen.
 15. The method of claim 14, further comprising advancing the penetrating tip to pierce through the occlusion.
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